

DMS — IV Sem : Solution of Model Question paper-I

Q 1) a) S.T the Compound proposition

$$[(p \leftrightarrow q) \wedge (q \leftrightarrow r) \wedge (r \leftrightarrow p)] \Leftrightarrow [(p \rightarrow q) \wedge (q \rightarrow r) \wedge (r \rightarrow p)]$$

for primitive statements p, q, r is logical equivalence

p	q	r	a $p \leftrightarrow q$	b $q \leftrightarrow r$	c $r \leftrightarrow p$	d $p \rightarrow q$	e $q \rightarrow r$	f $r \rightarrow p$	anbnc	dneft
0	0	0	1	1	1	1	1	1	1	1
0	0	1	1	0	0	1	1	0	0	0
0	1	0	0	0	1	1	0	1	0	0
0	1	1	0	1	0	1	1	0	0	0
1	0	0	0	1	0	0	1	1	0	0
1	0	1	0	0	1	0	1	1	0	0
1	1	0	1	0	0	1	0	1	0	0
1	1	1	1	1	1	1	1	1	1	1

∴ The above Compound proposition is logical equivalence.

b) Establish the validity of the following argument using the rules of inference:

$$\{p \wedge (p \rightarrow q) \wedge (s \vee r) \wedge (r \rightarrow \neg q)\} \rightarrow (s \vee t)$$

→ The given argument reads

$$\begin{array}{l} p \\ p \rightarrow q \\ s \vee r \\ r \rightarrow \neg q \\ \hline \therefore s \vee t \end{array}$$

we note that

$$(s \vee r) \wedge (r \rightarrow \neg q) \Leftrightarrow (\neg s \rightarrow r) \wedge (r \rightarrow \neg q)$$

$$\Rightarrow (\neg s \rightarrow \neg q), \text{ Syllogism}$$

$$\Leftrightarrow q \rightarrow s \text{ / Contrapositive}$$

$$\therefore (p \rightarrow q) \wedge (s \vee r) \wedge (r \rightarrow \neg q) \Rightarrow (p \rightarrow q) \wedge (q \rightarrow s)$$

$$\Rightarrow (p \rightarrow s), \text{ Syllogism}$$

Consequently,

$$P \wedge (P \rightarrow Q) \wedge (S \vee T) \wedge (T \rightarrow \sim Q) \Rightarrow P \wedge (P \rightarrow S)$$

$$\Rightarrow S \quad | \text{ Rule of detachment}$$

$$\Rightarrow S \vee T \quad | \text{ disjunctive Amplification}$$

© For the universe of all integers, let $p(x)$, $q(x)$, $r(x)$, $s(x)$ and $t(x)$ denote the following open statements.

$p(x): x > 0$, $q(x): x$ is even, $r(x): x$ is a perfect square, $s(x): x$ is divisible by 3, $t(x): x$ is divisible

by 7. Write the following statements in symbolic form.

(i) At least one integer is even

(ii) There must a positive integer that is even

(iii) If x is even, then x is not divisible by 3

(iv) No even integer is divisible by 7

(v) There must a even integer divisible by 3

Soln: (i) $\exists x, q(x)$

(ii) $\exists x [p(x) \wedge q(x)]$

(iii) $\forall x [q(x) \rightarrow \sim s(x)]$

(iv) $\forall x, \sim t(x)$

(v) $\exists x, [q(x) \wedge s(x)]$

(3)

2a) Define a tautology. prove that for any proposition;
 P, Q, R the compound propositions.

$\{(P \rightarrow Q) \wedge (Q \rightarrow R)\} \rightarrow \{P \rightarrow R\}$ is a tautology

Soln: A compound proposition which is true for all possible truth values of its components is called a tautology.

The Following truth table proves the result

P	Q	R	$P \rightarrow Q$	$Q \rightarrow R$	$(P \rightarrow Q) \wedge (Q \rightarrow R)$	$P \rightarrow R$	$((P \rightarrow Q) \wedge (Q \rightarrow R)) \rightarrow (P \rightarrow R)$
0	0	0	1	1	1	1	1
0	0	1	1	1	1	1	1
0	1	0	1	0	0	1	1
0	1	1	1	1	1	1	1
1	0	0	0	1	0	1	1
1	0	1	0	1	0	0	1
1	1	0	1	0	0	1	1
1	1	1	1	1	1	1	1

(b) Prove the following using laws of logic
 $P \rightarrow (Q \rightarrow R) \Leftrightarrow (P \wedge Q) \rightarrow R$

Soln: $P \rightarrow (Q \rightarrow R) \Leftrightarrow \sim P \vee (\sim Q \vee R)$ [Law of negation of a conditional]
 $\Leftrightarrow (\sim P \vee \sim Q) \vee R$ [Associative law]

$$\Leftrightarrow \sim(p \wedge q) \vee r$$

[De Morgan's law]

$$\Leftrightarrow (p \wedge q) \rightarrow r$$

[Law of negation of a conditional]

© Give (i) direct proof (ii) indirect proof

(iii) proof by contradiction for the following statement "If n is an odd integer then $n+9$ is an even integer".

Soln: Direct proof: Assume that n is an odd integer.

Then $n = 2k+1$, for some integer k .

Then $n+9 = 2k+1+9 = 2(k+5)$ which is even.

Indirect proof: Assume that $n+9$ is an odd integer,

$$\Rightarrow n+9 = 2k+1$$

$$\Rightarrow n = 2k-8 = 2(k-4)$$

which shows that n is even.

Thus if $n+9$ is not even, then n is not odd.

This proves the contrapositive of the given statement

(iii) proof by contradiction:

Assume that given statement is false. That is assume that n is odd and $n+9$ is odd.

$$\text{Since } n+9 \text{ is odd } n+9 = 2k+1 \Rightarrow n = 2k-8 = 2(k-4)$$

which shows that n is even. This contradicts the assumption that n is odd. Hence given statement is true

Module-2

③ a) Prove that $1^2 + 3^2 + 5^2 + \dots + (2n-1)^2 = \frac{n(n+1)(2n+1)}{3}$ by mathematical induction.

Soln: let $s(n)$ denote the given statement

Basis step: we note that $s(1)$ is the statement

$$1^2 = \frac{1}{3} \times 1 \times 3 = 1.$$

Therefore $s(1)$ is true.

Induction step: we assume that $s(k)$ is true for $n=k$, where $k \geq 1$. Then

$$1^2 + 3^2 + \dots + (2k-1)^2 = \frac{k(2k+1)(2k+1)}{3}$$

Adding $(2k+1)^2$ on both sides

$$1^2 + 3^2 + 5^2 + \dots + (2k-1)^2 + (2k+1)^2 = \frac{1}{3} k(2k+1)(2k+1) + (2k+1)^2$$

$$= \frac{1}{3} (2k+1) [k(2k+1) + 3(2k+1)]$$

$$= \frac{1}{3} (2k+1) (2k^2 + 5k + 3)$$

$$= \frac{1}{3} (2k+1) (k+1) (2k+3)$$

This is precisely the statement $s(k+1)$.

b) let $a_0 = 1, a_1 = 2, a_2 = 3$ and $a_n = a_{n-1} + a_{n-2} + a_{n-3}$ for $n \geq 3$. Prove that $a_n \leq 3^n, \forall n \in \mathbb{Z}^+$.

Soln: ~~$s(n) = a_0 \leq 3$~~ let $s(n) \equiv a_n \leq 3^n, n \geq 3$.

$s(0) \equiv a_0 \leq 3^0$ true $\therefore a_0 = 1$

$s(1) \equiv a_1 \leq 3^1$ true $\therefore a_1 = 2$

$$S(3) = S(3): a_2 < 3^2 \text{ true } \because a_2 = 3$$

Assume $S(0), S(1), S(2) \dots S(k)$ is true for $k \geq 2$

$$S(k+1): a_{k+1} < 3^{k+1}$$

$$a_{k+1} = a_k + a_{k-1} + a_{k-2} \text{ for } k \geq 2$$

$$< 3^k + 3^{k-1} + 3^{k-2}$$

$$\leq 3^k + 3^k + 3^k = 3^k(1+1+1) = 3^{k+1}$$

By mathematical induction $S(n)$ is true for $n \geq 3$.

© Find the number of ways of arrangement of the letters of the word 'TALLAHASSEE' which have no adjacent A's.

Soln: The number of letters in the given word is 11 of which 3 are A's, 2 each are L's, E's and S's and 1 each are T and H.

The number of arrangements (permutations) of the letters in the given word is

$$\frac{11!}{3! 2! 2! 2! 1! 1!} = 831600$$

If we disregard the A's the remaining 8 letters can be arranged in $\frac{8!}{2! 2! 2! 1! 1!} = 5040$

In each of these arrangements, there are 9 possible locations for the three A's. These locations can be chosen in $({}^9_3)$ ways. Therefore Number of arrangements having no adjacent A's is

$$5040 \times ({}^9_3) = 5040 \times \frac{9!}{3!6!} = 5040 \times 84 = 423360$$

OR

④ @ Determine the coefficient of xyz^2 in the expansion of $(2x-y-z)^4$

Soln: By the Multinomial theorem, we note that the general term in the expansion of $(2x-y-z)^4$ is

$$({}^4_{n_1 \ n_2 \ n_3}) (2x)^{n_1} (-y)^{n_2} (-z)^{n_3}$$

For $n_1=1, n_2=1, n_3=2$ this becomes

$$({}^4_{1 \ 1 \ 2}) (2x)(-y)(-z)^2 = ({}^4_{1 \ 1 \ 2}) x(2)(-1)x(-1)^2 yz^2$$

This shows that the required coefficient is

$$({}^4_{1 \ 1 \ 2}) \times (2)(-1) \times (-1)^2 = \frac{4!}{1!1!2!} \times (-2) = \underline{\underline{-24}}$$

- (8)
- (b) In how many ways one can distribute 8 identical marbles in 4 distinct containers so that (i) no container is empty (ii) The fourth container has an odd number of marbles in it.

Soln: (i) First we distribute one ball to each container. Then we distribute the remaining 4 balls into 4 containers.

The No. of ways $C(4+4-1, 4) = C(7, 4) = 35$.

- (ii) Suppose we put one ball into it. Then remaining 7 balls can be distributed into the remaining 3 containers in $C(3+7-1, 7) = C(9, 7)$ ways

Suppose we put 3 balls into it

$$\Rightarrow C(3+5-1, 5) = C(7, 5) \text{ ways.}$$

Suppose we put 5 balls into it

$$\Rightarrow C(3+3-1, 3) = C(5, 3) \text{ ways}$$

Suppose we put 7 balls into it

$$\Rightarrow C(3+1-1, 1) = C(3, 1) \text{ ways}$$

$$\text{The total no. of ways} = 35 + 21 + 10 + 3 = 70$$

- (c) How many positive integers n can be formed using the digits 3, 4, 4, 5, 5, 6, 7 if we want to exceed 5,00,000?

Soln: Here n must be of the form

$$n = x_1 x_2 x_3 x_4 x_5 x_6 x_7.$$

Where $x_1, x_2, \dots, x_7$ are the given digits with $x_1 = 5, 6$ or 7 . Suppose we take $x_1 = 5$. Then $x_2, x_3, x_4, x_5, x_6, x_7$ is an arrangement of the remaining 6 digits which contains two 4's and one each 3, 5, 6, 7. The number of such arrangements is

$$\frac{6!}{1! 2! 1! 1! 1! 1!} = 360$$

Similarly $x_1 = 6$, The number of arrangement is

$$\frac{6!}{1! 2! 2! 1! 1!} = 180.$$

Similarly $x_1 = 7$ The number of arrangement is

$$\frac{6!}{1! 2! 2! 1! 1!} = 180$$

By sum rule, number of n 's of the designed type

$$\text{is } 360 + 180 + 180 = \underline{720}$$

Module-3

⑤ a) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = \begin{cases} 3x-5 & \text{if } x > 0 \\ 1-3x & \text{if } x \leq 0 \end{cases}$

Find $f^{-1}(0)$, $f^{-1}(1)$, $f^{-1}(-1)$, $f^{-1}(3)$, $f^{-1}(6)$, $f^{-1}([-5, 5])$ and $f^{-1}([-5, 5])$.

Soln: ~~$f^{-1}(0)$~~ We find that $f(x) = 0$ only when $x = 5/3$ (observe that $f(x) \neq 0$ for $x \leq 0$) Therefore

$$f^{-1}(0) = \{5/3\}.$$

$$f^{-1}(1) = \{x \in \mathbb{R} \mid f(x) = 1\} = \{2, 0\}$$

$$f^{-1}(-1) = \{x \in \mathbb{R} \mid f(x) = -1\} = \{4/3\}$$

$$f^{-1}(3) = \{x \in \mathbb{R} \mid f(x) = 3\} = \{8/3, -2/3\}$$

$$f^{-1}(6) = \{x \in \mathbb{R} \mid f(x) = 6\} = \{11/3\}$$

$$f^{-1}([-5, 5]) = \{x \in \mathbb{R} \mid f(x) \in [-5, 5]\}$$

$$= \{x \in \mathbb{R} \mid -5 \leq f(x) \leq 5\}$$

When $x > 0$, we have $f(x) = 3x-5$, Therefore

$$-5 \leq f(x) \leq 5 \text{ whenever } -5 \leq (3x-5) \leq 5,$$

$$\text{or } 0 \leq 3x \leq 10 \text{ or } 0 < x \leq 10/3$$

When $x \leq 0$ we have $f(x) = -3x+1$, Therefore

$$-5 \leq f(x) \leq 5 \text{ whenever } -5 \leq (-3x+1) \leq 5 \text{ or}$$

$$-6 \leq -3x \leq 4 \text{ or } 2 \geq x \geq -4/3 \text{ or } -4/3 \leq x \leq 2.$$

$$\text{Thus } f^{-1}([-5, 5]) = \{x \in \mathbb{R} \mid -4/3 < x < 2 \text{ or } 0 < x < 10/3\} \\ = \{ -4/3, 10/3 \}$$

Similarly we find that

$$\begin{aligned} f^{-1}([-6, 5]) &= \{x \in \mathbb{R} \mid f(x) \in [-6, 5]\} \\ &= \{x \in \mathbb{R} \mid -6 \leq f(x) \leq 5\} \end{aligned}$$

When $x > 0$, we have $f(x) = 3x - 5$, Therefore.

$$-6 \leq (3x - 5) \leq 5, \text{ whenever } -1 \leq 3x \leq 10. \text{ or}$$

$$-1/3 \leq x \leq 10/3.$$

When $x \leq 0$, we have $f(x) = 1 - 3x$, Therefore

$$-6 \leq (1 - 3x) \leq 5 \text{ whenever } -7 \leq -3x \leq 4.$$

$$\text{or } -7/3 \leq x \leq 4/3. \quad 7/3 \geq x \geq -4/3 \text{ or}$$

$$-4/3 \leq x \leq 7/3.$$

$$\text{Thus } f^{-1}([-6, 5]) = \{x \in \mathbb{R} \mid -4/3 \leq x \leq 7/3, \text{ or } -1/3 \leq x \leq 10/3\}$$

$$\therefore f^{-1}([-6, 5]) = \underline{\underline{(-4/3, 10/3)}}$$

5) State pigeon hole principle. Prove that any set of 29 persons at least 5 persons have been born on the same day of the week.

Soln: Treating 7 days of a week as 7 pigeonholes and 29 persons as pigeons, we find by using generalized pigeonhole principle that at least one day of the week is assigned to $(\frac{29-1}{7}) + 1 = 5$ or more pigeons. In other words at least 5 of

any 29 persons must have been born on the same day of the week.

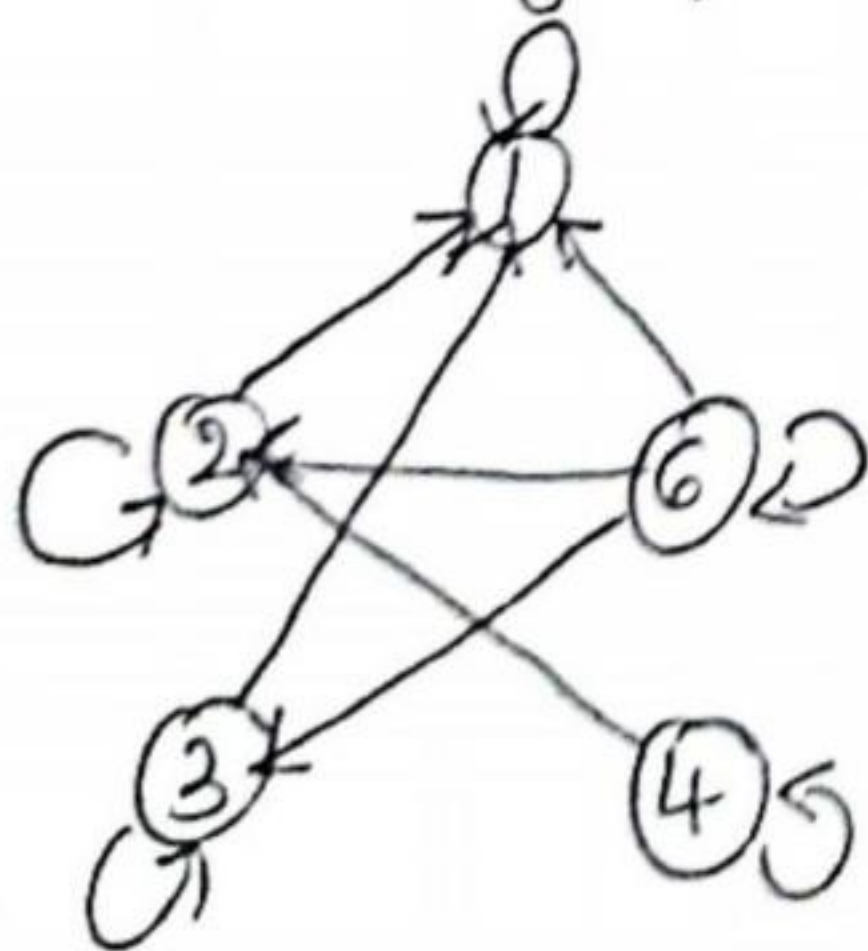
© let $A = \{1, 2, 3, 4, 6\}$ and R be a relation on 'A' defined by aRb iff "a is a multiple of b" represent the relation 'R' as a matrix, draw the digraph of Relation R.

Soln: From the definition of the given R we note that

$$R = \{(1, 1), (2, 1), (2, 2), (3, 1), (3, 3), (4, 1), (4, 2), (4, 4), (6, 1), (6, 2), (6, 3), (6, 6)\}$$

$$M_R = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 1 \end{bmatrix}$$

The digraph of R is as shown below



⑬
 (a) If $f: A \rightarrow B$, $g: B \rightarrow C$, $h: C \rightarrow D$ are three functions
 then prove that $h \circ (g \circ f) = (h \circ g) \circ f$.

Soln: We first note that both $(h \circ g) \circ f$ and $h \circ (g \circ f)$
 are functions from A to D

For any $x \in A$ we have

$$\begin{aligned} [(h \circ g) \circ f](x) &= (h \circ g)[f(x)] = h \circ g(y) \text{ where } y = f(x) \\ &= h[g(y)] = \underset{\text{①}}{h(z)}, \text{ where } z = g(y) \end{aligned}$$

$$\begin{aligned} \text{and } [h \circ (g \circ f)](x) &= h[g \circ f(x)] \\ &= h[g(f(x))] = h[g(y)] = h(z) - \text{②} \end{aligned}$$

Results ① & ② show that

$$[(h \circ g) \circ f](x) = [h \circ (g \circ f)](x) \text{ for } x \in A$$

Therefore $(h \circ g) \circ f = h \circ (g \circ f)$ and theorem is proved

(b) Show that if $n+1$ numbers are chosen from 1 to $2n$
 then at least one pair add to $2n+1$

Soln: $A_1 = \{1, 2n\}$ $A_2 = \{2, 2n-1\}$.. $A_{n-1} = \{n-1, n+2\}$.

$$A_n = \{n, n+1\}$$

These are the only sets containing two numbers from
 1 to $2n$ whose sum is $2n+1$

Since every number from 1 to $2n$ belongs to one

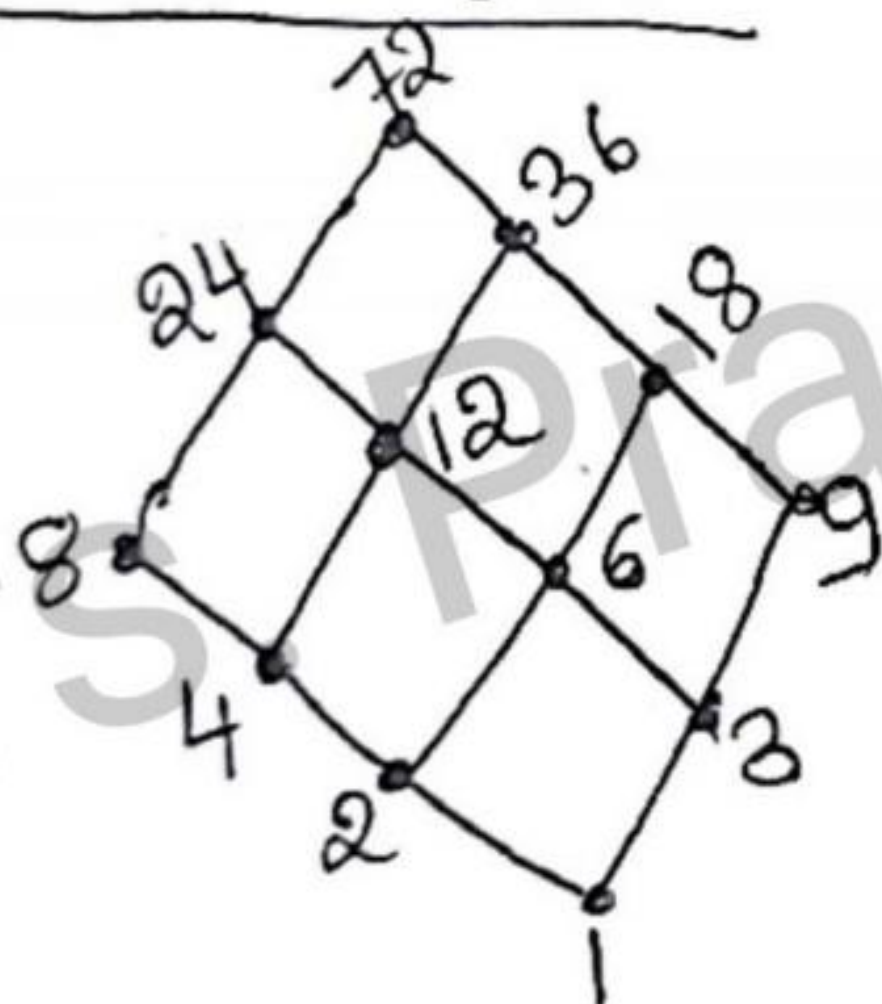
of the above sets, each of the $n+1$ numbers chosen must belong to one of the sets. Since there are only n sets, two of the $n+1$ chosen numbers have to belong to the same set. These two numbers have their sum equal to $2n+1$.

© Draw the Hasse diagram representing the positive divisors of 72.

Soln: positive divisors of 72 can be written as

$$D_{72} = \{1, 2, 3, 4, 6, 8, 9, 12, 18, 24, 36, 72\}$$

Hasse Diagram:



(F) @ In how many ways the 26 letters of English alphabet can be permuted so that none of the patterns CAR, DOG, PUN or BYTE occurs?

Soln: Let S denote the set of all permutations of 26 letters.
Then $|S| = 26!$

Let A_1 be the set of all permutations in which CAR appears. This word CAR consists of three letters which form a single block. The set A_1 therefore consists of all permutations which contain this single block and the 23 remaining letters.

Therefore $|A_1| = 24!$

Similarly if A_2, A_3, A_4 are the sets of all permutations which contain DOG, PUN and BYTE respectively.

We have $|A_2| = 24!, |A_3| = 24!, |A_4| = 23!$

likewise we find that

$$|A_1 \cap A_2| = |A_1 \cap A_3| = |A_2 \cap A_3| = (26 - 6 + 2)! = 22!$$

$$|A_1 \cap A_4| = |A_2 \cap A_4| = |A_3 \cap A_4| = (26 - 7 + 2)! = 21!$$

$$|A_1 \cap A_2 \cap A_3| = (26 - 9 + 3)! = 20!$$

$$|A_1 \cap A_2 \cap A_4| = |A_1 \cap A_3 \cap A_4| = |A_2 \cap A_3 \cap A_4| = (26 - 10 + 3)! = 19!$$

$$|A_1 \cap A_2 \cap A_3 \cap A_4| = (26 - 13 + 4)! = 17!$$

Required Number of permutations is Given by

$$|\bar{A}_1 \cap \bar{A}_2 \cap \bar{A}_3 \cap \bar{A}_4| = |S| - \sum |A_i| + \sum |A_i \cap A_j| - \sum |A_i \cap A_j \cap A_k|$$

$$+ |A_1 \cap A_2 \cap A_3|$$

$$= 261 - (3 \times 241 + 231) + (3 \times 22! + 3 \times 21!) + (201 + 3 \times 19!) + 171$$

⑤ Define Derangements. In how many ways can each of 10 people select a left glove and right glove out of a total of 10 pairs gloves so that no person selects a matching pair of gloves?

Soln: A permutation of n distinct objects in which none of the objects is in its natural place is called a derangement.

Required number is

$$D_{10} = 10! \left[1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \frac{1}{5!} + \dots + \frac{1}{10!} \right]$$

$$\approx \underline{\underline{10 \cdot e^{-1}}}$$

© Solve the recurrence relation, $C_n = 3C_{n-1} - 2C_{n-2}$
for $n \geq 2$, given $C_1 = 5$, $C_0 = 3$

Soln: Here the coefficients of C_n, C_{n-1}, C_{n-2} are respectively, $C_n = 1$, $C_{n-1} = -3$, & $C_{n-2} = 2$

Therefore the characteristic equation is

$$k^2 - 3k + 2 = 0 \text{ or } (k-2)(k-1) = 0.$$

whose roots are $k_1 = 2$ and $k_2 = 1$ Therefore the General solution for a_n is

$$a_n = A \times 2^n + B \times 1^n \quad \text{--- (1)}$$

where A & B are arbitrary constants. Using the given conditions $C_1 = 5$ and $C_0 = 3$ in this we

$$\text{get } 5 = 2A + B \text{ and } 3 = 4A + B$$

solving these we get $A = -1$ & $B = 7$ putting these into (1) we get

$$\boxed{a_n = -2^n + 7}$$

Q. In how many ways 5 number of a's, 4 number of b's and 3 number c's can be arranged so that all the identical letters are not in a single block?

Soln: The given letters are $5+4+3=12$ in number, of which 5 are a's 4 are b's and 3 are c's. If S is the set of all permutations of these letters we have

$$|S| = \frac{12!}{5!4!3!}$$

Let A_1 be the set of arrangements of letters where the 5 a's are in single block. The number of such arrangements is $|A_1| = \frac{8!}{4!3!}$

Similarly if A_2 is the set of arrangements where the 4 b's are in a single block and A_3 is the set of arrangements where the 3 c's are in a single block we have $|A_2| = \frac{9!}{5!3!}$ and $|A_3| = \frac{10!}{5!4!}$

likewise the required number of arrangements is $|\bar{A}_1 \cap \bar{A}_2 \cap \bar{A}_3| = |S| - \{|A_1| + |A_2| + |A_3|\} + \{|A_1 \cap A_2| + |A_1 \cap A_3| + |A_2 \cap A_3|\} - |A_1 \cap A_2 \cap A_3|$

$$= \frac{12!}{5!4!3!} - \left\{ \frac{8!}{4!3!} + \frac{9!}{5!3!} + \frac{10!}{5!4!} \right\} + \left\{ \frac{5!}{3!} + \frac{6!}{4!} + \frac{7!}{5!} \right\} - 3!$$

$$= 27720 - (280 + 504 + 1260) + (20 + 30 + 42) - 6$$
$$= \underline{\underline{25762}}$$

- ⑧ In how many ways can we arrange the letters of the word CORRESPONDENTS so that there are
- exactly 2 pairs of consecutive identical letters?
 - at least 3 pairs of consecutive identical letters?
 - no pair of consecutive identical letters?

Soln: In the word CORRESPONDENTS, there occur one each of C, P, D and T, and two of each of O, R, E, S, N. If S is the set of all permutations of these 14 letters we have

$$|S| = \frac{14!}{(2!)^5}$$

Let A_1, A_2, A_3, A_4, A_5 be the sets of permutations in which O's, R's, E's, S's, N's appear in pairs respectively. Then $|A_i| = \frac{13!}{(2!)^4}$ for $i=1, 2, 3, 4, 5$

$$|A_i \cap A_j| = \frac{12!}{(2!)^3}, \quad |A_i \cap A_j \cap A_k| = \frac{11!}{(2!)^2}$$

$$|A_i \cap A_j \cap A_k \cap A_l| = \frac{10!}{(2!)^2},$$

$$|A_1 \cap A_2 \cap A_3 \dots \cap A_5| = 9!$$

From these we get

$$S_0 = N = |S| = \frac{14!}{(2!)^5}$$

$$S_1 = C(5, 1) \times \frac{13!}{(2!)^4}$$

$$S_2 = C(5, 2) \times \frac{12!}{(2!)^3}$$

$$S_3 = C(5, 3) \times \frac{11!}{(2!)^2}$$

$$S_4 = C(5, 4) \times \frac{10!}{2!}$$

$$S_5 = C(5, 5) \times 9!$$

Accordingly the number of permutations where there is no pair of identical letters is

$$E_0 = s_0 - \binom{1}{1} s_1 + \binom{2}{2} s_2 - \binom{3}{3} s_3 + \binom{4}{4} s_4 - \binom{5}{5} s_5$$

$$= \frac{141}{(2!)^5} 5 - \binom{5}{1} \times \frac{13!}{(2!)^4} + \frac{5}{2} \times \frac{12!}{(2!)^3} - \binom{5}{3} \times \frac{11!}{(2!)^2} + \binom{5}{4} \times \frac{10!}{2!} - \binom{5}{5} \times 9!$$

Next the number of permutations where there are exactly two pairs of consecutive identical letters is

$$E_2 = s_2 - \binom{3}{1} s_3 + \binom{4}{2} s_4 - \binom{5}{3} s_5$$

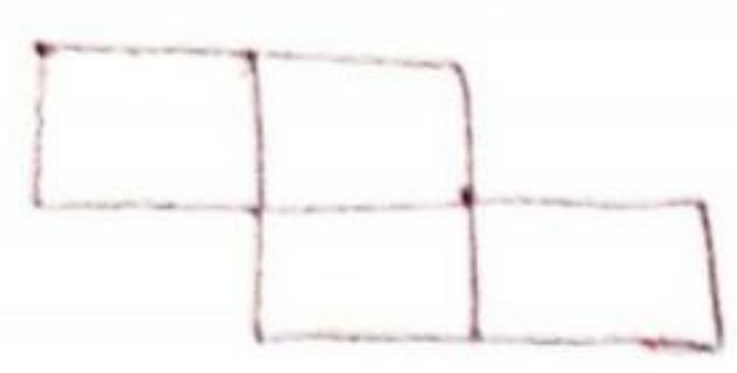
$$= \binom{5}{2} \times \frac{12!}{(2!)^3} - \binom{3}{1} \binom{5}{3} \times \frac{11!}{(2!)^2} + \binom{4}{2} \binom{5}{4} \times \frac{10!}{2!} - \binom{5}{3} \binom{5}{5} \times 9!$$

lastly the number of permutations where there are at least three pairs of consecutive identical letters is

$$h_3 = s_3 - \binom{3}{2} s_4 + \binom{4}{2} s_5$$

$$= \binom{5}{3} \times \frac{11!}{(2!)^2} - \binom{3}{2} \binom{5}{4} \times \frac{10!}{2!} + \binom{4}{2} \binom{5}{5} \times 9!$$

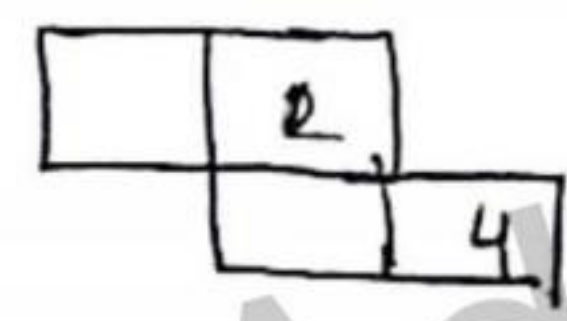
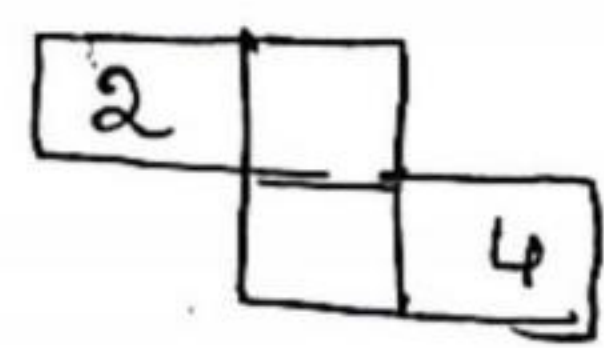
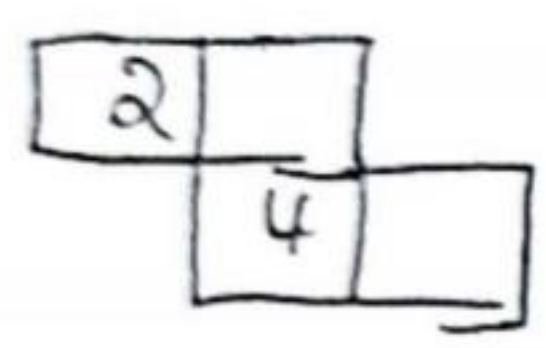
② Find the rook polynomial for the chess board as shown in the figure



Soln: $r_0 = r_1x + r_2x^2 + r_3x^3 + r_4x^4$

Maximum two rows $\Rightarrow r_3 = r_4 = 0$

By inspection $r_2 = \text{no. of rows one can place two nonthreatening rooks} = 3$



$\therefore R = 1 + 4x + 3x^2$

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⑥ Solve the recurrence relation $a_{n+2} - 3a_{n+1} + 2a_n = 0$

$$a_0 = 1, a_1 = 6$$

soln: characteristic equation is

$$r^2 - 3r + 2 = 0$$

roots are $r = 1, 2$.

$$\text{as } a_n = k_1(1)^n + k_2(2)^n$$

$$\text{put } n = 0, \quad a_0 = k_1 + k_2$$

$$n = 1, \quad a_1 = 6 = k_1 + 2k_2$$

$$\therefore \underline{a_n = -4 + 5(2)^n}$$

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Module-5

Q. 11 If H, K are subgroups of a group G , prove that $H \cap K$ is also a subgroup of G . Is $H \cup K$ a subgroup of G ?

Soln: Let G be a group and H, K be subgroups of G .

Take any $a, b \in H \cap K$, Then $a, b \in H$ and $a, b \in K$.

Therefore $ab^{-1} \in H$ and $ab^{-1} \in K$ because H & K are subgroups of G . This means that $ab^{-1} \in H \cap K$.

Hence $H \cap K$ is a subgroup of G .

Thus Intersection of two subgroups of a group G is a subgroup of G .

also Union of two subgroups of a group need not be a subgroup.

Q. Define Klein-4 group. Verify $A = \{1, 3, 5, 7\}$ is a Klein 4 group.

Soln: Klein four group is an abelian group with 4 elements, in which each element is self inverse and in which composing of two of the three non identity elements produces the third one.

If the set forms $A = \{1, 3, 5, 7\}$ forms a Klein group. mainly we need to check the following properties.

(1) closure property: For any two elements $a, b \in A$

the product ab is also in A .

2. Inverse property: Every element a in A has an inverse a^{-1} in A such that $aa^{-1} = e$, where e is the identity element.

3. Associativity: The operation defined on A which is multiplication modulo 8 is associative.

Closure:

$1 \times 1 \equiv 1 \pmod{8}$	$3 \times 3 \equiv 9 \equiv 1 \pmod{8}$
$1 \times 3 \equiv 3 \pmod{8}$	$3 \times 5 \equiv 15 \equiv 7 \pmod{8}$
$1 \times 5 \equiv 5 \pmod{8}$	$3 \times 7 \equiv 21 \equiv 5 \pmod{8}$
$1 \times 7 \equiv 7 \pmod{8}$	$5 \times 5 \equiv 25 \equiv 1 \pmod{8}$
	$5 \times 7 \equiv 35 \equiv 3 \pmod{8}$
	$7 \times 7 \equiv 49 \equiv 1 \pmod{8}$

Each pair of elements multiplied modulo 8 gives us another element in A .

Hence A is closed under multiplication modulo 8.

Inverse:

$1 \times 1 = 1$	so $1(1)^{-1} = 1$
$3 \times 3 = 1$	so $3(3)^{-1} = 3$
$5 \times 5 = 1$	so $5(5)^{-1} = 5$
$7 \times 7 = 1$	so $7(7)^{-1} = 7$

Each element in A has an inverse within A .

© state and prove Lagrange's theorem

(25)

Soln: Statement: If G is a finite group and H is a subgroup of G , then the order of H divides order of G .

Proof: Since G is finite, H is finite. Therefore the number of cosets of H in G is finite. Let $Ha_1, Ha_2, Ha_3, \dots, Ha_n$ be the distinct right cosets of H in G . Then by the right coset decomposition of G we

$$G = Ha_1 \cup Ha_2 \cup \dots \cup Ha_n$$

$$\text{So that } o(G) = o(Ha_1) + o(Ha_2) + \dots + o(Ha_n)$$

$$\text{But } o(Ha_1) = o(Ha_2) = \dots = o(H).$$

$$o(G) = n o(H)$$

This shows that $o(H)$ divides $o(G)$

OR

- (10) (a) Show that (i) the identity of G is unique
(ii) The inverse of each element of G is unique.

Soln: (i). Suppose e and e' be two identity elements of G , $e, e' \in G$

Suppose e is the identity element, then

$$e * e' = e \quad - (1)$$

Suppose e' is the identity element then

$$e * e' = e' \quad - (2)$$

on comparing $\underline{e = e'}$

Hence identity element is unique.

(ii). Suppose b & c are two inverse of a , let $a \in G$.

$$b * a = a * b = e \quad \text{--- (I)}$$

$$c * a = a * c = e \quad \text{--- (II)}$$

Since Associativity property holds Good in group.

$$(b * a) * c = b * (a * c) = b * e = b. \quad e * c = c \quad \text{--- (1)}$$

$$b * (a * c) = b * e = b \quad \text{--- (2)}$$

From (1) & (2) we can say $\boxed{b = c}$

Hence inverse of a is unique.

(b) Show that $(A, \cdot)$ is an abelian group where $A = \{a \in \mathbb{R} \mid a \neq -1\}$ and for any $a, b \in A$,

$$a \cdot b = a + b + ab.$$

Soln: When $a \neq -1$, and $b \neq -1$, we note that $a + b + ab \neq -1$, Therefore $\cdot$ is a binary operation on G . ~~The~~ For any $a, b, c \in G$ we have

$$a(b \cdot c) = a.(b + c + bc)$$

$$= a + (b + c + bc) + (ab + ac + abc)$$

$$= [(a + b + ab) + c] + (a + b + ab)c$$

$$= (a + b + ab) \cdot c$$

$$= (ab) \cdot c.$$

$$a(b \cdot c) = (ab) \cdot c.$$

Thus $\cdot$ is associative.

For any $a \in G$, we have $a \cdot 0 = 0 \cdot a = a$

$$a \cdot 0 = a + 0 + a \cdot 0 = a = 0 + a + 0 \cdot a = 0 \cdot a$$

Thus 0 is the identity element in G under $+$.

For any $a \in G$, we put $a' = -1/(1+a)$ then $a' \in G$ and

$$a \cdot a' = a + a' + aa' = a - \frac{a}{(1+a)} + \frac{a^2}{(1+a)} = 0 = a' \cdot a$$

Thus $a' = -1/(1+a) \in G$ is the inverse of a under $\cdot$.

Thus $(G, \cdot)$ is an abelian group.

© Let $G = S_4$, for $d = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix}$ find the subgroup

$H = \langle d \rangle$. Determine the left cosets of H in G .

soln: We know that S_4 is the multiplicative group of all permutations of the set $A = \{1, 2, 3, 4\}$. The identity element in this group is the identity permutation

$$P_0 = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 1 & 2 & 3 & 4 \end{pmatrix}$$

The cyclic group generated by given d is

$$\langle d \rangle = \{d, d^2, d^3, \dots, d^n = P_0\}$$

$$d^2 = d \cdot d = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix}$$

$$d^3 = d^2 \cdot d = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix}$$

$$d^4 = d^3 \cdot d = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 1 & 2 & 3 & 4 \end{pmatrix} = P_0$$

$$H = \langle \alpha \rangle = \{ \alpha, \alpha^2, \alpha^3, \alpha^4 = p_0 \}$$

We note that $|H| = 4$, Also $|G| = |S_4| = 24$

Number of left coset of H in G is $[G:H] = \frac{|G|}{|H|}$

$$= \frac{24}{4} = \underline{\underline{6}}$$

Left coset of H in G is 6

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